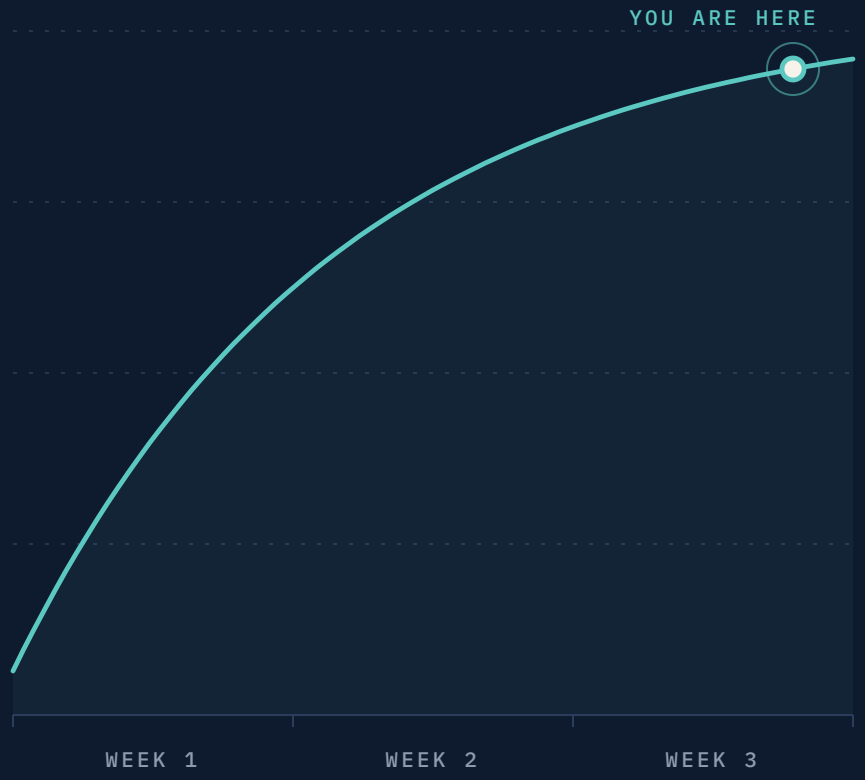
WEEK 3 · DAY 4 · OVERVIEW

The Daily Market Snapshot



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The Desk Lead's Note

Before every morning meeting I want one page. Where the curve and our two spread series closed on the last market date, how they moved over a day, a week and a month, and where each sits in its own year. The curve's shape. What the day's curve move does to the September book, as a what-if. A web page I can open and a workbook I can check in Excel. Then make it one command for any date, set it to run every morning, and keep the script in our record, never its pages. A page for 30 September shows 30 September, even when the file runs on.

ABOUT 50 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

Your Route Through Day 4

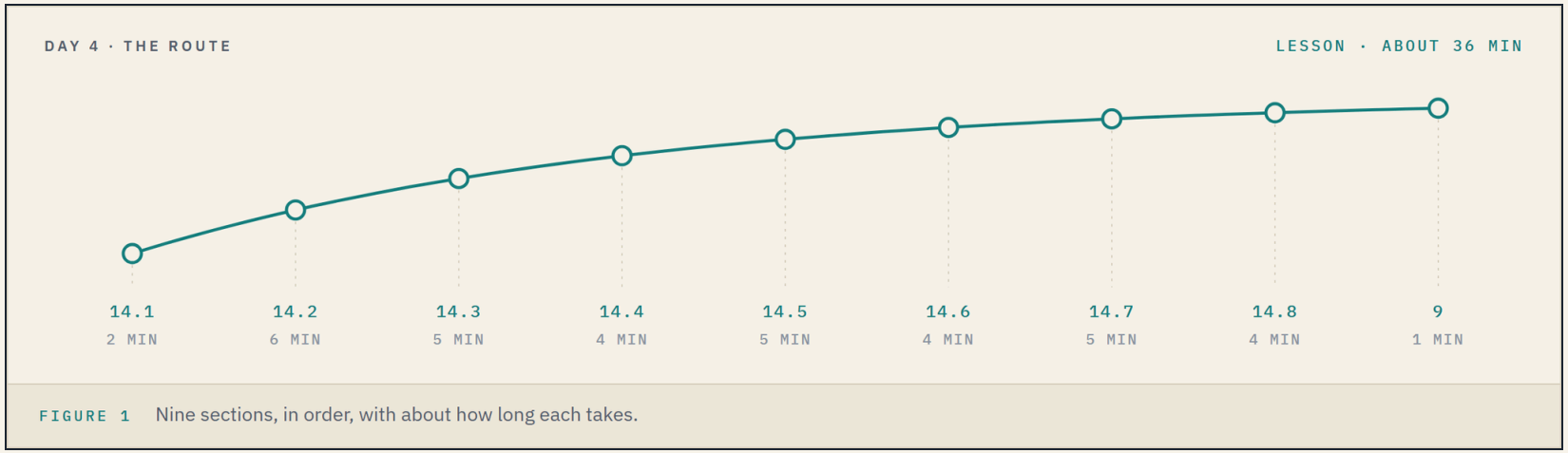


FIGURE 1 Nine sections, in order, with about how long each takes.

	SECTION	ABOUT
14.1	One Page, One Date, One Rule	2 min
14.2	**`snapshot_tables`**, Docstring First	6 min
14.3	The Book Section: DV01 and a What-If	5 min
14.4	Only Rows On or Before the Date	4 min
14.5	Two Charts and the Page	5 min

	SECTION	ABOUT
14.6	The Workbook, Opened in Excel	4 min
14.7	**`snapshot.py`**: One Command, Any Date	5 min
14.8	Every Morning, and the Record	4 min
9	Bring It Home	1 min

Time. About 36 minutes for the lesson, then about 12 for the four exercises.

14.1 One Page, One Date, One Rule

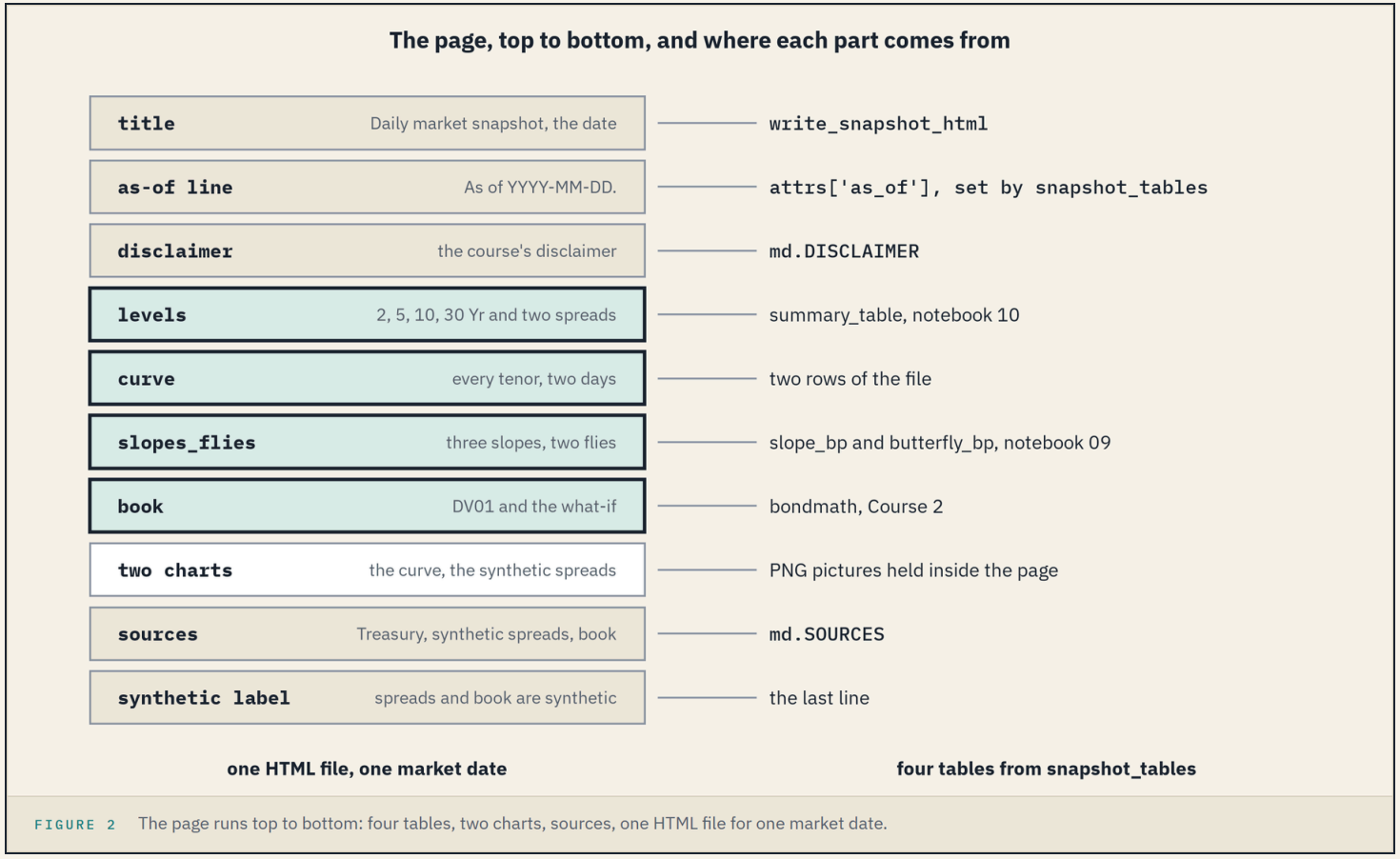


FIGURE 2 The page runs top to bottom: four tables, two charts, sources, one HTML file for one market date.

THE RULE

DESCRIBES	NEVER
was, rose, fell, over a day, a week, a month	will, expect, should
a z-score over 252 rows, today included	stretched, due to revert
the share of the window at or below today	cheap, rich, a signal
what the book would do if the curve moved so, labelled a what-if	what the book will do

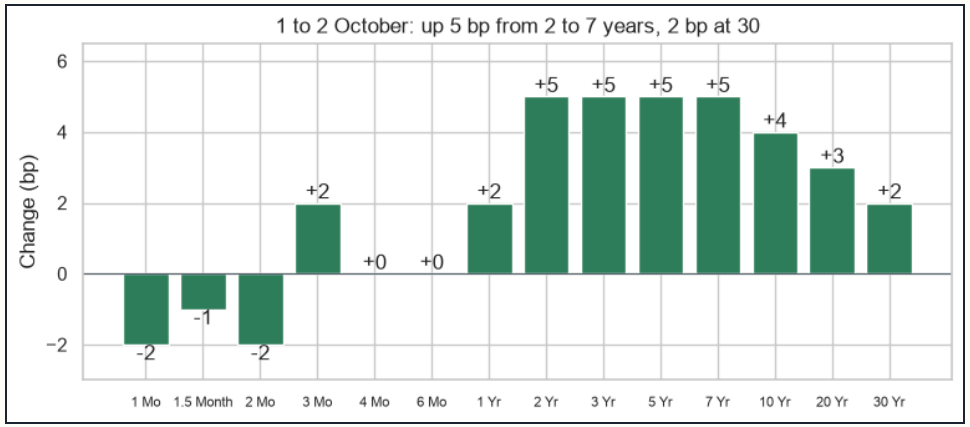
One date, four tables, two charts, and one rule: every number says where something was, never where it goes.

14.2 snapshot_tables, Docstring First

THE LEVELS TABLE, 2 OCTOBER

SERIES	LEVEL	CHG_1D_BP	ZSCORE	PCT_RANK
2 Yr	4.83	5.00	2.51	98.02
5 Yr	5.06	5.00	2.80	99.60
10 Yr	5.28	4.00	3.08	99.60
30 Yr	5.63	2.00	3.08	99.60

THE CURVE'S MOVE BY TENOR



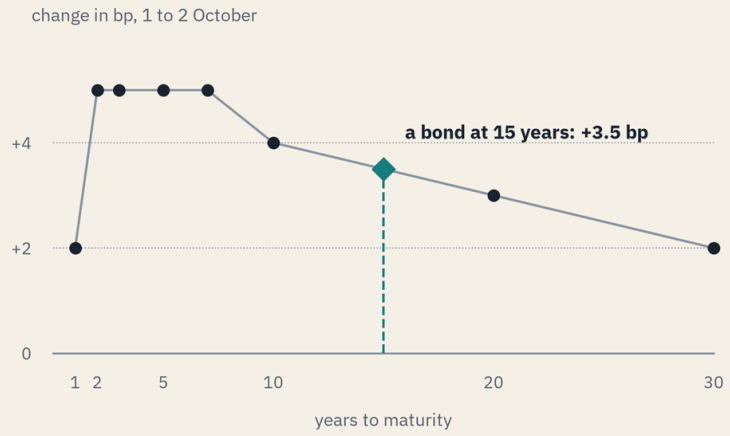
EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Z-score	`=STANDARDIZE(A252,AVERAGE(A1:A252),STDEV.S(A1:A252))`	3.075234
Percentile rank	`=COUNTIF(A1:A252,"<="&A252)/COUNT(A1:A252)*100`	99.603175

One call builds three of the page's tables for 2 October, and the 10 Yr's z-score ties to Excel's within 4.6e-13.

14.3 The Book Section: DV01 and a What-If

A what-if along the curve: each bond moves by the day's change at its maturity



`interpolate_yield`
the change at the bond's maturity

`dirty_price, twice`
at its yield, then at yield + the change

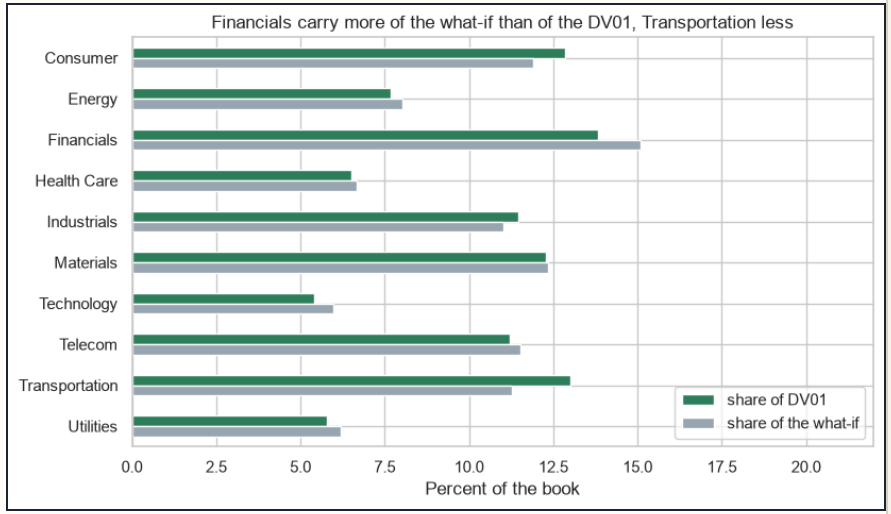
(after - before) x face / 100
the bond's change, in dollars

summed
by sector, by rating bucket, the book

a what-if: only the Treasury curve moves, no spread change, no carry; it says what the book would do, never what it will

FIGURE 3 The curve rose 2 to 5 bp by maturity; each bond moves by its own point, and the dollars sum by sector.

WHERE IT LANDS



83 percent of Transportation's DV01 sits beyond 10 years, where the curve rose only 2 to 4 bp.

14.4 Only Rows On or Before the Date

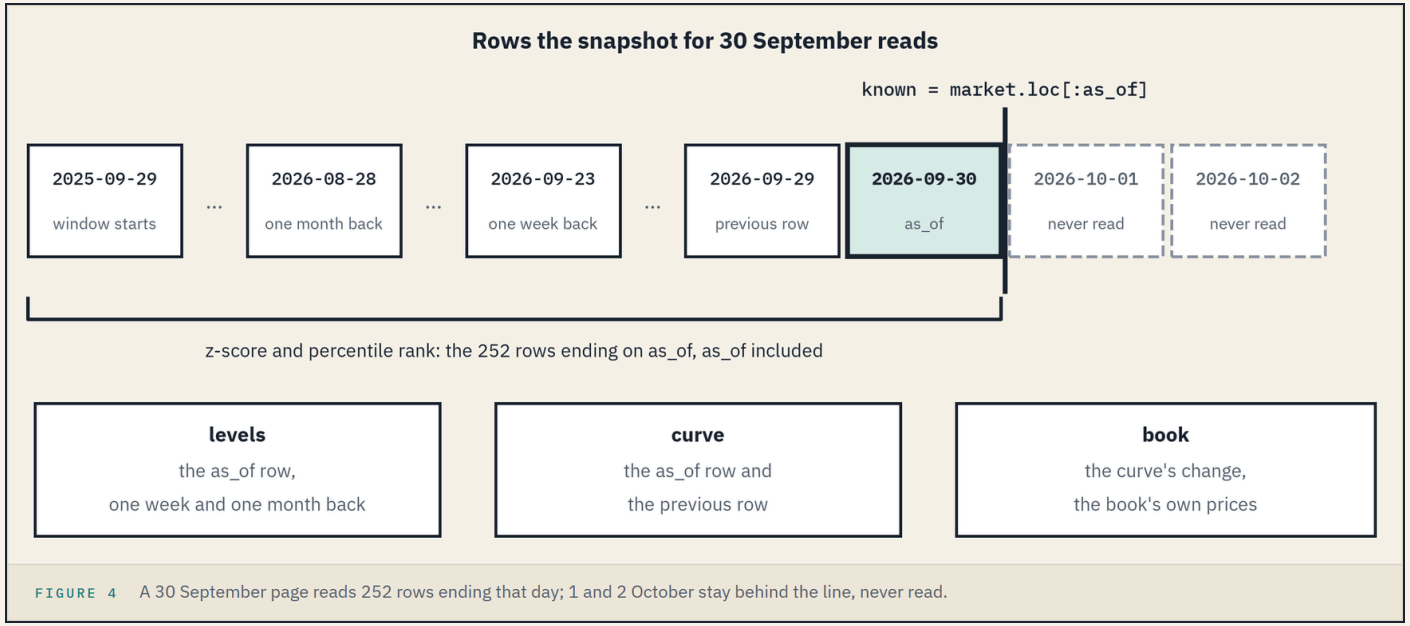


FIGURE 4 A 30 September page reads 252 rows ending that day; 1 and 2 October stay behind the line, never read.

CUT AND UNCUT

```
four tables equal; the 10 Yr on 30 September: 5.29
```

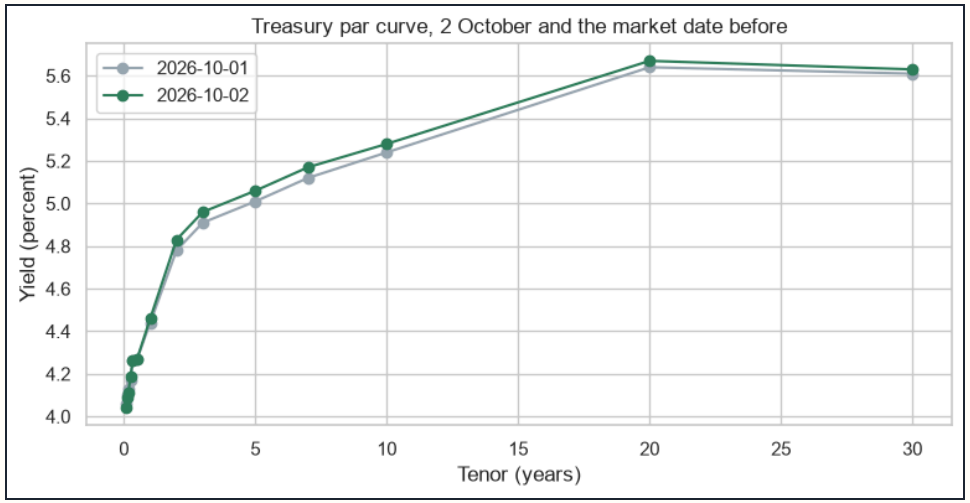
A DATE WITH NO ROW

```
ValueError: 2026-09-07 is not a market date: market has no row for it
```

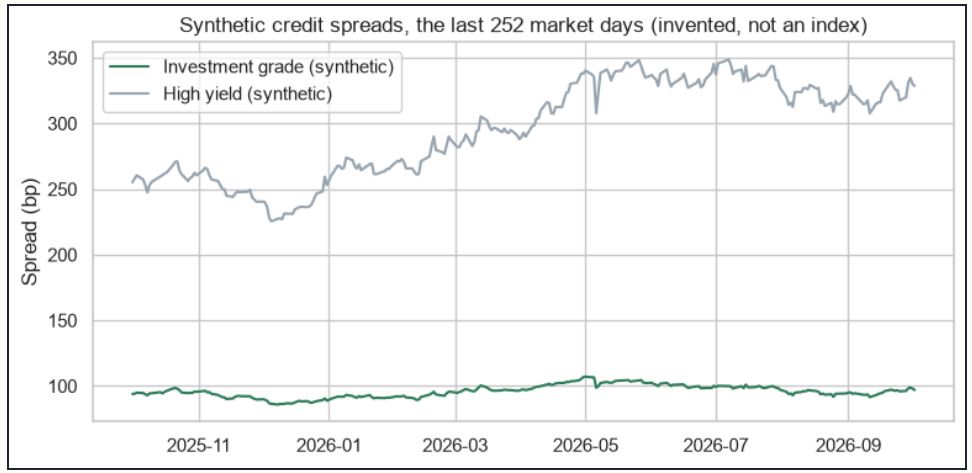
The snapshot refuses a date the market never had and never reads past its date, and two tests now hold both.

14.5 Two Charts and the Page

THE CURVE ON BOTH DAYS



THE SPREADS, 252 DAYS



THE PAGE, CHECKED AS TEXT

```
7 of 7 checks pass
```

One function writes one self-contained file, title to sources, and seven plain checks prove what is on it.

14.6 The Workbook, Opened in Excel

EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
The sheets	the tabs	**about**, **levels**, **curve**, **slopes_flies**, **book**
The curve's change	`=(D2-C2)*100` in F, filled down	within 8.9e-16 bp of column E
DV01 by sector	`=SUMIFS(G2:G19,B2:B19,"sector")`	306996.110402
The what-if, both ways	`=SUMIFS(H2:H19,B2:B19,"sector")`, then `"rating bucket"`	-1193297.89

READ IT BACK, SHEET BY SHEET

```
['about', 'levels', 'curve', 'slopes_flies', 'book'] as of 2026-10-02
{'about': 4, 'levels': 6, 'curve': 14, 'slopes_flies': 5, 'book': 18}
```

The workbook opens in Excel with its date on the first sheet, and Excel's own `SUMIFS` gives the book's DV01 back to the sixth decimal.

14.7 snapshot.py: One Command, Any Date

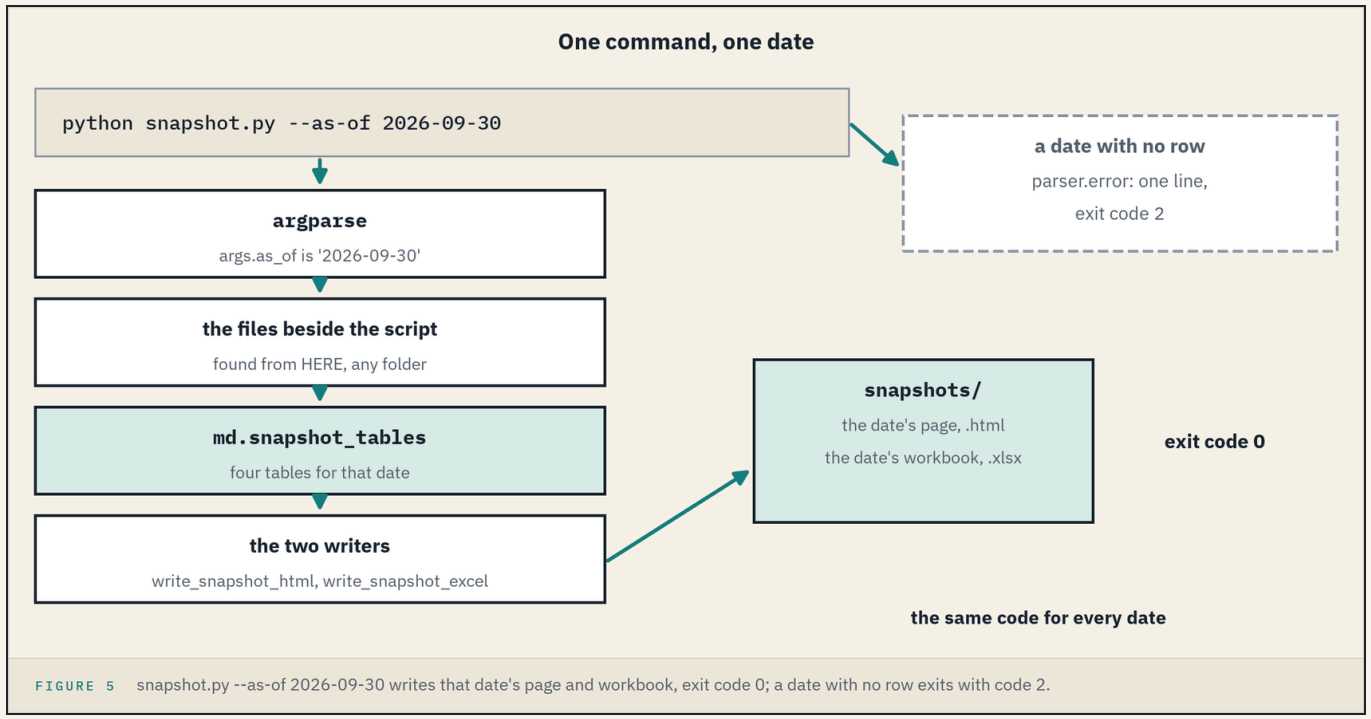


FIGURE 5 snapshot.py --as-of 2026-09-30 writes that date's page and workbook, exit code 0; a date with no row exits with code 2.

ITS USAGE

```
usage: snapshot.py [-h] [--as-of AS_OF]
options:
  -h, --help     show this help message and exit
```

ITS EXIT CODE

```
exit code 2
```

A DATE WITH NO ROW, THROUGH THE SCRIPT

```
snapshot.py: error: 2026-09-07 is not a market date: market has no row for it
```

14.8 Every Morning, and the Record

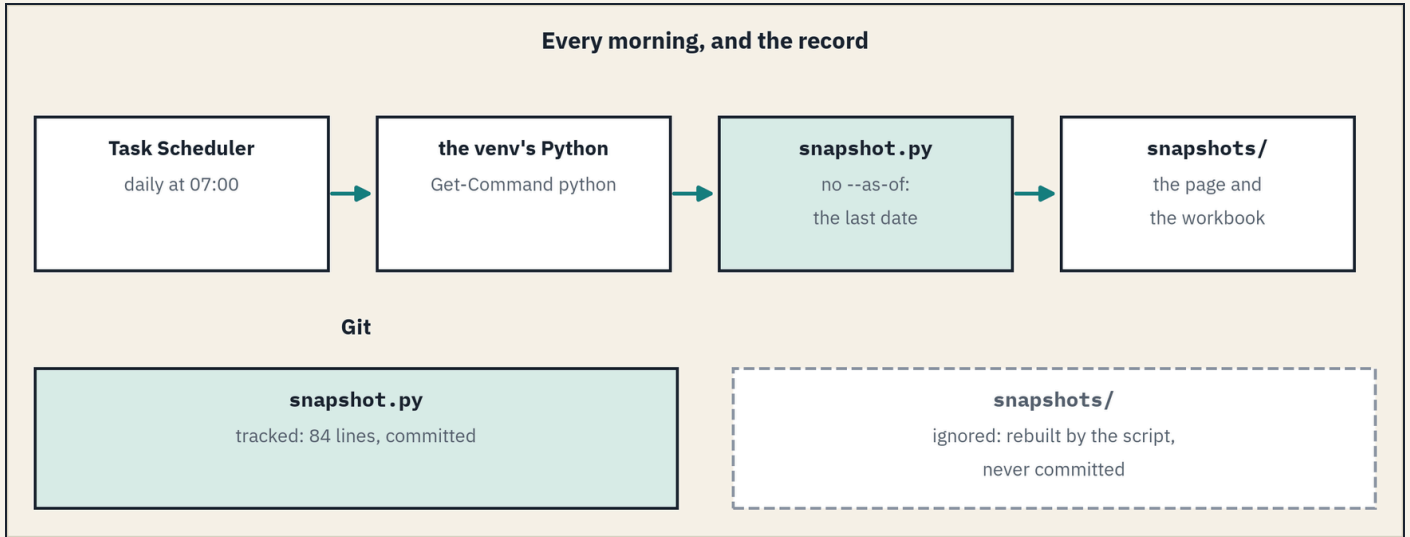


FIGURE 6 Task Scheduler runs snapshot.py each morning at 07:00; the script is committed and the snapshots are ignored.

WHAT GIT SEES

```
!git -C "{work}" status --short
```

```
M marketdata.py
 M test_marketdata.py
?? snapshot.py
?? snapshots/
```

THE TASK, AS TEXT FOR YOUR OWN MACHINE

```
schtasks /Create /TN "pfi_daily_snapshot" /SC DAILY /ST 07:00 /TR "'$python' '$HOME\projects\my_bondmath\snapshot.py'"
schtasks /Query /TN "pfi_daily_snapshot"
```

Task Scheduler can run the command every morning, and Git keeps the 84 lines that write the pages and none of the pages.

Copy the module and script home.

Bring It Home

IN A POWERSHELL TERMINAL, WITH THE COURSE'S .VENV ACTIVE

```
# the work folder: use the name your setup cell printed
$work = "$env:TEMP\pfi_course3_14_k4w7q2xr"
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$work\marketdata.py", "$work\test_marketdata.py", "$work\snapshot.py" .
Copy-Item "$work\course3_synthetic_spreads.csv" .
Add-Content .gitignore "snapshots/"
python -m pytest -q test_marketdata.py
python snapshot.py
Invoke-Item snapshots\snapshot_2026-10-02.html
git status --short
git add .gitignore marketdata.py test_marketdata.py snapshot.py
git commit -m "Notebook 14: snapshot_tables, write_snapshot_html, write_snapshot_excel, and snapshot.py"
```

pytest shows 55 passed, and the page opens in your browser.

You Can Now

- Build a one-page snapshot's tables for any market date from rows on or before it only.
- Put the book's DV01 by sector and rating beside a what-if of the day's curve move, labelled as one.
- Write a self-contained HTML page and an Excel workbook, and check both as data.
- Prove a date-driven report reads nothing after its date.
- Turn notebook code into a script with **`argparse`**, run it for any date, and read its exit code.
- Schedule a script with Task Scheduler and keep its outputs out of Git.

NEXT

WEEK 3 · DAY 5

Course 3 Capstone

A vendor's extract proved against its note and the market's own calendar, cleaned by one function, reconciled, appended, and turned into the snapshot for two dates. You write the code.

